Productivity J-Curve

Value dips before it climbs

PROFESSIONAL DEFINITION

The Productivity J-Curve describes how a general-purpose technology first depresses measured productivity — as firms invest in intangible complements — before delivering a steep, delayed payoff.

WHO DEVELOPED IT

Formalised by Erik Brynjolfsson, Daniel Rock and Chad Syverson (2021), building on Brynjolfsson's long study of the IT productivity paradox.

WHY IT WAS DEVELOPED

It was developed to explain why transformative technologies appear to fail in the statistics for years, reassuring decision-makers that the dip is part of the path.

FIGURE 1 · PRODUCTIVITY J-CURVE

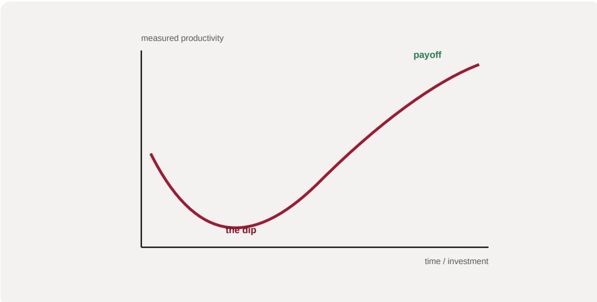

Figure 1. A schematic of the Productivity J-Curve as applied in BARBM312 (illustrative).

HOW IT IS USED

An AI programme's measured productivity is read against the curve; investment in complementary intangibles is tracked to confirm a slope is being built.

WHEN IT IS USED

When boards lose patience with AI ROI, and when distinguishing a genuine lag from a genuine absence of value.

BY WHOM IT IS USED

CFOs, transformation leaders and investment committees.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It supplies a rigorous, evidence-based reason not to abandon AI programmes at the bottom of the dip, reframing the cost of complements as investment, not failure.

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- **Adoption → Impact Lag** · Slibrary 5
- **Intangible Capital Iceberg** · Slibrary 5
- **Diffusion S-Curve** · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E., Rock, D. and Syverson, C. (2021) 'The Productivity J-Curve', American Economic Journal: Macroeconomics, 13(1), pp. 333–372.

Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age. New York: Norton.